NEXUS PATENT APPLICATION - USPTO READY

UNITED STATES PATENT AND TRADEMARK OFFICE

NON-PROVISIONAL PATENT APPLICATION

35 U.S.C. § 111(a) - Conversion from Provisional

APPLICATION DATA SHEET

Field	Value
Filing Type	Non-Provisional (Conversion)
Provisional Filed	November 27, 2025
Conversion Deadline	November 27, 2026
Entity Status	Micro Entity
Applicant/Inventor	Kevin Caracozza
Address	321 Lyndhurst Avenue, Lyndhurst, NJ 07071
Email	kcaracozza@gmail.com
Phone	551-795-5862
Docket Number	NEXUS-2026-001

APPLICATION TITLE

UNIVERSAL SMALL COLLECTIBLES RECOGNITION AND MANAGEMENT SYSTEM WITH ENCLOSED ROBOTIC SCANNING CHAMBER, ADAPTIVE INTELLIGENCE, AND CROSS-CATEGORY LEARNING

CLAIM COUNT AND USPTO FEES (Micro Entity)

Component	Calculation	Amount
Total Claims	85	-
Independent Claims	34	-
Excess Claims (over 20)	65 × $20	$1,300
Excess Independent (over 3)	31 × $42	$1,302
Base Filing Fee	-	$320
TOTAL USPTO FEE		**$2,922**

CLASSIFICATION

- **Primary:** G06K 9/00 - Pattern Recognition
- **Secondary:** G06K 9/62 - Recognition methods
- **Additional:** G06N 3/08 - Neural Networks
- **Additional:** G06Q 30/02 - Marketing/Price Estimation
- **Additional:** B25J 9/16 - Robotic manipulators
- **Additional:** G06T 7/00 - Image analysis
- **Additional:** G06V 30/10 - Optical character recognition

FIELD OF THE INVENTION

The present invention relates to automated recognition and management systems for collectible items. The system comprises an enclosed scanning chamber with controlled lighting, an articulated robotic arm with camera positioning, adaptive OCR, cross-category AI learning, and distributed cloud architecture for recognizing, cataloging, grading, and managing small collectibles including trading cards, sports cards, coins, stamps, comic books, action figures, video games, and vinyl records.

Total Addressable Market: $248+ billion global collectibles market

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BACKGROUND - PROBLEMS SOLVED

2.1 Fragmented Solutions

Prior art systems are category-specific (TCGPlayer for TCGs, Discogs for vinyl, COMC for sports cards). Shop owners need 5+ separate systems.

2.2 Manual Data Entry

- Error rates: 5-15% in item identification
- Time cost: 30-60 seconds per item
- Cannot scale operations

2.3 Poor Recognition Accuracy

- PSA Genamint: ~85% accuracy
- DelverLens: MTG only
- Smartphone apps: Variable lighting, inconsistent results

2.4 No Cross-Category Learning

Learning in one category provides zero benefit to others.

2.5 Privacy Concerns

Cloud-only solutions expose sensitive business data.

SUMMARY OF INNOVATIONS

Hardware Components

1. **Enclosed scanning chamber** - opaque walls, controlled environment
2. **LED lighting system** - 3000K-6500K color temp, 0-100% intensity
3. **Articulated robotic arm** - 3+ DOF, camera positioning
4. **High-resolution camera** - 12MP+ with macro capability

Software Components

1. **Category detection** - automatic item type identification
2. **Multi-region extraction** - category-specific OCR regions
3. **Bulletproof OCR** - 100% uptime, graceful degradation
4. **Cross-category AI** - patterns transfer between categories
5. **Visual fingerprinting** - perceptual hash matching
6. **Adaptive shop personality** - UI evolves per installation

Architecture Components

1. **Privacy-first** - sensitive data stays local
2. **Marketplace integration** - 20+ platforms
3. **Universal database** - all categories, one schema

TECHNICAL IMPROVEMENTS

Metric	Prior Art	NEXUS
Recognition Accuracy	60-85%	99%+
Items/Hour	10-15	100+
System Uptime	Variable	100%
Categories	1	Universal
Cross-Category Learning	None	Yes

KEY INNOVATIONS (12 Novel Elements)

#	Innovation	Claims
1	Enclosed scanning chamber	1, 16, 51, 55-56
2	Robotic arm positioning	2, 17, 38, 53, 57
3	Multi-region extraction	3, 18, 34, 45
4	Bulletproof OCR	4, 19
5	Cross-category AI	5, 20, 32, 39-42
6	Adaptive shop personality	6, 21, 37
7	Privacy-first architecture	7, 22, 35, 47, 83-84
8	Consumer handheld	8, 23, 52, 58-61
9	Barcode-visual validation	9, 24, 36, 48
10	AI grading prediction	10, 25, 33, 43-44
11	Visual fingerprinting	12, 27, 66, 68-69
12	Zero-sort call numbers	73-76

ZERO-SORT INNOVATION (Strongest Novel Claim)

The Core Insight: Call numbers have NO relationship to item attributes.

Traditional Physical Sorting (Competitors)

- Card scanned → Sorted into bin by attribute
- Requires bins for each category
- Cannot query by attributes not binned

NEXUS Zero-Sort

- Card scanned → Assigned next sequential call number
- Placed in next physical position (regardless of value)
- Database handles ALL organization virtually

Example:

- AA-0001: Black Lotus ($500,000)
- AA-0002: Forest basic land ($0.05)
- AA-0003: Mickey Mantle ($12.6M)

All stored sequentially. Query by ANY attribute returns locations.

No prior art combines: OCR identification + sequential call numbers + database-indexed location tracking + elimination of physical sorting.

CLAIM SUMMARY (85 Total)

System Claims 1-30 (15 independent, 15 dependent)

- Enclosed chamber, robotic arm, multi-region extraction
- Bulletproof OCR, cross-category AI, adaptive personality
- Privacy architecture, handheld device, barcode integration
- AI grading, shop intelligence, visual fingerprints
- Universal database, position profiles, marketplace

Method Claims 31-50 (8 independent, 12 dependent)

- Universal scanning method, cross-category learning
- Grading prediction, multi-region extraction
- Privacy sync, barcode-visual validation
- Adaptive UI, robotic capture

Apparatus Claims 51-65 (4 independent, 11 dependent)

- Scanning chamber apparatus, handheld scanner
- Robotic arm apparatus, lighting apparatus

AI/ML Claims 66-71 (2 independent, 4 dependent)

- Visual fingerprint AI, shop intelligence AI

New Claims 72-85 (6 independent, 8 dependent)

- RL training (2M steps)
- Zero-sort call number system
- Marketplace integration
- Multi-node distributed processing
- Privacy enforcement

PRIOR ART DIFFERENTIATION

US20040002387A1 (ABANDONED)

Card reader/scanner by Daniel Grady (2002)

- **Status:** ABANDONED - No licensing required
- **Missing:** Zero-sort, location tracking, marketplace, robotic arm

US20230325392A1 (TAG Grading - ACTIVE)

TAG's OCR patent family (2023)

- **Focus:** GRADING (condition assessment)
- **NEXUS Focus:** CATALOGING (identification, inventory)
- **Distinction:** Different purpose entirely

Commercial Competitors

Competitor	Approach	NEXUS Differentiation
CardMill	Physical sorting	Zero-sort eliminates
Roca	Physical bins	Database-indexed
SortSwift	Mechanical sort	No moving parts

SUPPORTED CATEGORIES (16+)

1. Trading Card Games (MTG, Pokemon, Yu-Gi-Oh, etc.)
2. Sports Cards (Topps, Panini, Upper Deck)
3. Sports Memorabilia (signed items, game-used)
4. Entertainment Memorabilia (autographs, props)
5. Historical Memorabilia (documents, signatures)
6. Coins and Numismatics (US, world, ancient)
7. Stamps and Philately
8. Comic Books
9. Action Figures and Toys
10. Diecast Models and Vehicles
11. Model Trains
12. Model Kits and Miniatures
13. LEGO and Building Sets
14. Video Games
15. Physical Media (CD, DVD, Blu-ray)
16. Vinyl Records

HARDWARE ARCHITECTURE

Three-Node Distributed System

Node 1: BROK (192.168.1.169)

- Raspberry Pi 5 (8GB RAM)
- Google Coral Edge TPU
- Local OCR, database, API server

Node 2: SNARF (192.168.1.172)

- Raspberry Pi 5 (4GB RAM)
- Dual OwlSight cameras + CZUR scanner
- Hardware control, video streaming

Node 3: ZULTAN (192.168.1.152)

- Raspberry Pi 5 (8GB RAM) or x86 server
- Marketplace sync, heavy ML tasks
- Optional GPU acceleration

Communication: ZeroMQ with CurveZMQ encryption

REINFORCEMENT LEARNING (Claim 72)

Position profiles optimized through **2,000,000 training steps**

Training Environment

- Simulated scanning chamber
- Virtual collectibles (varying sizes, reflectivity)
- Reward: image quality, OCR confidence, accuracy

Training Parameters

- Algorithm: PPO (Proximal Policy Optimization)
- Episodes: ~50,000 scanning scenarios
- Convergence: 99%+ accuracy on validation

Results

- Non-obvious optimal angles discovered
- 3-5% OCR accuracy improvement
- Automatic glare reduction for reflective surfaces

MARKETPLACE INTEGRATION (Claims 77-79)

Supported Platforms (20+)

- **TCGs:** TCGPlayer, eBay, CardMarket
- **Sports:** COMC, MySlabs, Sportlots
- **Comics:** MyComicShop, Atomic Avenue
- **Coins:** Heritage Auctions, APMEX
- **Stamps:** HipStamp, Delcampe
- **Vinyl:** Discogs, Reverb LP

Features

- Unified listing creation (one-click multi-platform)
- Cross-platform price monitoring
- Automatic repricing engine
- Inventory sync (prevents overselling)
- Unified order management

VERIFIED MARKET DATA

Segment	2024 Value	Source
Total Collectibles	$462.82B	UniDatos
Trading Cards	$21.4B	Market Decipher
Sports Memorabilia	$33.6B	Market Decipher
Coins	$18.1B	Transparency MR
Comics	$17.0B	IMARC Group

Competitor Valuations

- TCGPlayer: $295M acquisition (eBay, 2022)
- PSA/Collectors Holdings: $4.3B valuation

Fee Comparison

- TCGPlayer: 10.25% commission
- eBay: 13.25% FVF

DECLARATION AND CERTIFICATION

I hereby declare that:

1. I am the original inventor of the subject matter described
2. I have reviewed and understand the contents of this application
3. This application claims benefit of Provisional App filed Nov 27, 2025
4. I acknowledge the duty to disclose material information

Micro Entity Certification (37 CFR 1.29):

- Fewer than 4 prior patent applications
- Gross income below 3× median household income
- No obligation to assign to non-qualifying entity

Inventor: Kevin Caracozza

Residence: Lyndhurst, New Jersey, USA

Mailing Address: 321 Lyndhurst Avenue, Lyndhurst, NJ 07071

Citizenship: United States of America

Date: _____

Signature: _____

ATTORNEY INSTRUCTIONS

Documents Provided

1. Complete specification with detailed description
2. All 85 claims with proper dependency
3. Prior art analysis and differentiation
4. 21 figure descriptions for formal drawings

Action Items

1. Review claims for antecedent basis and format
2. Review specification for adequate claim support
3. Prepare formal drawings from descriptions
4. File non-provisional converting from provisional
5. Consider PCT filing (same deadline)

USPTO Fees (Micro Entity)

- Base filing: $320
- Excess claims: $1,300
- Excess independent: $1,302
- **Total: $2,922**

Contact

- **Email:** kcaracozza@gmail.com
- **Phone:** 551-795-5862

DOCUMENT STATUS

Metric	Value
Full Document	E:\NEXUS_USPTO_READY_PATENT_APPLICATION.md
Total Lines	4,126
Total Claims	85
Independent	34
Dependent	51
USPTO Fee	$2,922
Status	**READY FOR ATTORNEY REVIEW**

This PDF is a summary. Full specification with all claims available in the markdown file.

File: E:\NEXUS_USPTO_READY_PATENT_APPLICATION.md (151 KB)